

Explainable AI-Based Interview Coaching System

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Abstract- Technical interviews have been used to evaluate candidates for software engineering, data science, and other technical positions for many years. These interviews are used to evaluate technical knowledge as well as communication skills, clarity, confidence, and so on. Many candidates will struggle to present their ideas clearly because they are nervous and/or hesitate, or because the way they present their answers are not structured. Therefore, there is a need for intelligent systems that assist candidates in preparing for interviews.

This research proposes an Explainable AI Interview Coach that will evaluate responses to interview question(s) using a dual-analysis framework. The Explainable AI Interview Coach will generate appropriate interview questions based on the candidate's resume or the selected domain using a large language model. The candidate can respond to the interview question by providing a text response or an audio response. An audio response will be transcribed into a text response for semantic evaluation and various acoustic features such as pitch variation, speech rate, number of pauses, jitter, shimmer, and total number of filler words will be extracted from the audio for information on confidence. Those acoustic features will be used for predicting a confidence score using an XGBoost model with a resulting R^2 value of 0.932.

Keywords: Explainable AI, Interview Coaching System, Speech Analysis, Natural Language Processing, Confidence Prediction, Machine Learning, XGBoost, Speech Feature Extraction.

I. INTRODUCTION

Technical interviews are used to assess candidates for software engineering, artificial intelligence, and other technical roles. Technical interviews are meant to measure a candidate's ability to solve problems, technical capability, analytical thinking, and communication skills [1]. In addition to assessing logical ability, the interviewer also considers how well the candidate communicates their ideas and solutions to questions, as well as how they respond to follow-up questions (e.g., through clear speech and structured explanations). With regard to the perception of performance, a candidate's level of confidence, clarity of speech, and ability to explain themselves in a structured manner can have a significant impact on their performance during a technical interview [2].

Many applicants find it difficult to perform well in job interviews, even when they possess sufficient technical knowledge. Nervousness, uncertainty, inability to speak clearly, and lack of practice in interviews are frequently cited as reasons for poor performance [3]. Mock interviews with colleagues or mentors may not provide consistent and objective feedback. Due to this need for reliable feedback on interview performance and evaluation of responses from the perspective of an interviewer, there is an increased demand for automated interview environments that will simulate an actual interview and have structured measures for evaluating the responses of candidates [4]. Recent advances in artificial intelligence technology have produced intelligent systems that analyze both the responses of candidates (textual data)

and the speaking patterns of candidates (speech data) [5]. These intelligent systems have utilized natural language processing (NLP) methodologies to analyze the response of candidates for semantic correctness and have employed speech processing algorithms to identify the acoustic features related to speaking behaviour. Machine learning models will then analyze these acoustic and behavioural features to determine features such as hesitation, fluency of speech, and self-confidence.

An Explainable AI-Based Interview Coaching System will be developed to help technical applicants prepare for job interviews by generating interview questions from either a user's resume or from their preferred type of engineering job (domain) using a large language model (LLM) to create the question (based on the user's resume/directory) and provide candidates with the opportunity to answer via text and voice. The AI system will assess whether the candidate's response to the interview question is correct based on LLM evaluation and how well the candidate performed when verbally responding to the interview question using automated speech analysis [6]. The AI's LLM assessment and automated speech analysis will provide a complete assessment of the candidate's technical knowledge level as well as a complete assessment of their communication skillset. Therefore, the AI will provide the candidate with Explaining Feedback that will assist in identifying their key skillsets, while also identifying the areas in which they need to improve.

II. LITERATURE SURVEY

2.1 Automated Interview Preparation Systems

Automated systems for preparing for interviews have become increasingly popular as the use of AI in hiring and recruitment grows [7]. There have been numerous studies published that investigate the creation of intelligent platforms to automatically generate interview questions and evaluate the responses given by candidates. Most of these systems use natural language processing techniques to analyze candidate responses to written questions and provide automated feedback to candidates [8]. Most early research concentrated on generating structured interview question types and evaluating candidate responses based solely on keyword matches and semantic similarity to those keywords.

Numerous studies have recently made improvements in this field by incorporating the use of larger-sized language models capable of interpreting contextual information from both the resume and job description documents. These language models will produce more realistic-sounding interview questions that more closely resemble real-life hiring situations [9] therefore, candidates can practice answering domain-specific types of questions that relate directly to their background and experience in their chosen profession. While most automated systems for preparing for interviews are focused primarily on the right answers to questions, most of these same systems neglect the assessment of behavioural factors, e.g., how confidently candidates speak and how effectively they deliver their answers.

2.2 Speech Analysis for Confidence Detection

Analysis of the speech signal has been used extensively in behavioural computing to identify the level of confidence and emotional state during communication [10]. Acoustic characteristics used to evaluate the patterns of speakers include pitch variation, speech rate, pause duration, and energy levels; Research demonstrates that speakers who have confidence are typically characterized by stable pitch variation and displaying similar speech rates via fewer pauses where they may be hesitating before continuing speaking.

A number of studies have explored the ability of speech signal processing techniques to identify emotional states (e.g., stress, nervousness, and confidence) through the analysis of extracted acoustic features from the digital processing of speech signal recordings, using statistics or machine learning based methods [11]. Jitter and shimmer are highly relevant in analyzing the stability of voices and the use of filler-words and brief pause frequency would be relevant to determining if they are hesitating or showing a lack of confidence.

2.3 Machine Learning for Speech-Based Behavioural Analysis

Using machine learning algorithms to analyze speech has helped identify behavioural patterns based on the acoustic properties of the voice. The early days of using statistical models to classify the various properties of speech are becoming less common, as more and more speech classification has become reliant on supervised learning algorithms trained on labelled data sets of speech. The models trained through supervised learning have learned to establish a relationship between the acoustic properties of speech and the behaviours that can be inferred from them (e.g., confidence, emotion, fluency, etc.).

A recent study has demonstrated the use of ensemble learning techniques like gradient boosting algorithms to model the complex relationships between speech attributes and behavioural indicators. In particular, the model XGBoost has demonstrated the ability to make highly accurate predictions and provide feature importance analysis, which aids in obtaining explainability within the machine learning system [12]. Additionally, it has been shown that by incorporating natural language processing with speech analysis can lead to creating the ability to fully assess the content and delivery of the speech being evaluated.

III. METHODOLOGY

3.1 System Overview

The proposed Interview Coaching System has been built using a modular architecture that divides functions into defined modules (i.e., question generator, response evaluator, speaker evaluator/analysis, confidence score prediction). Users have the ability to either provide a copy of their resume or select an area for which they would like to prepare for a technical interview, and based on this selection, a large language model will provide the users with interview questions that will simulate actual technical job interview scenarios.

When responding to questions, candidates can provide their responses through either text or audio. If provided in audio format, the response will first be transcribed by a Speech Recognition Model into text before being assessed by a Language Model to determine whether the candidate's answer correctly fulfils the requirements set forth in the question. In parallel, the candidate's speech characteristics are extracted from the original audio signal and evaluated against a previously trained Machine Learning Model that will identify the candidate's Confidence Score.

3.2 Data Preparation

A synthetic dataset that contains 8,982 total samples was produced to supplement the limited availability of labelled audio interview data. Each of the audio samples was formed using statistically realistic distributions of 9 (nine) acoustic feature(s). Audio sample labels were produced using a hybrid method: initial labelling was based on an algorithmic formula then secondary labelling was to capture more holistic interactions between the features using LLM (GPT-3.5/Llama 3 Re) [13].

3.3 Speech Feature Extraction

A vital component of the proposed system for determining a candidate's level of confidence and fluency is the speech feature extraction process (or "speech features"). Acoustic features can be obtained from audio recordings using a variety of speech processing libraries including, but not limited to: Librosa, and Whisper [14], [15]. These extracted acoustic features will consist of many individual features including:

- **Pitch Mean** - Average frequency of speech over a given duration of time. Used to determine if the candidate's vocal tone has remained stable.
- **Pitch Variance** - Measures how much variation in pitch occurs throughout a candidate's entire speech sample.
- **Speech Rate** - Measures how fast or slow a candidate delivers their speech.
- **Pause Count** - Measures how many times a candidate pauses during their entire speech sample.
- **Average Pause Duration** - Average length of the candidate's pauses.
- **Energy Mean** - Measures how loud or intense a candidate speaks.
- **Jitter** - Measures how much a candidate's voice varies in pitch during his/her delivery.
- **Shimmer** - Measures how much amplitude (or loudness) a candidate's voice changes during speech.
- **Filler Words** - Counts the number of times a candidate used filler words, such as "um" and "uh", in their speech.

Collectively, these features will be used as inputs to the confidence prediction model by providing valuable insight into a candidate's overall speech performance.

3.4 Question Generation and Answer Evaluation

The system generates interview questions from a candidate's resume using a large language model. Those questions will

simulate realistic technical interviews, and can be classified into Easy, Medium, and Hard difficulty levels.

The candidate receives a response to their answers from the system through the use of Natural Language Processing techniques. By using the language model to examine the context and semantic meaning of a response, the system will determine whether or not the candidate answered the question correctly. In this way, candidates will receive constructive feedback based upon their communication skills and technical expertise.

3.5 Confidence Prediction Model

In this study, four different predictive modelling approaches were explored - Ridge regression, Random Forests, Multi-layer Perceptron neural network, and XGBoost (eXtreme Gradient Boosting). The model that showed the best prediction capabilities based on the highest Mean Absolute Error (MAE) and R^2 values was the XGBoost model [16]. In addition, the model built with the XGBoost algorithm, which is an ensemble learning method using gradient boosting decision trees, was found to be more computationally efficient as well as having better MAE and R^2 scores compared to the other models.

XGBoost has become a very popular algorithm for structured data due to its predictive performance as well as its ability to model complex interactions between features [17]. With XGBoost, the extracted speech feature set will be used as independent variables to train the model, which learns the relationship between the independent variables and confidence labels in the training data, and then produce a prediction of the confidence score for new speech inputs. In essence, the predicted confidence score gives an indication of how confident the interviewer was while responding to the question.

3.6 System Workflow Description

System Architecture

The architectural diagram depicts how different modules of the Interview Coaching System communicate with each other. A User interacts with the system through a User Interface (UI) used to upload resumes (or select topics for interview preparation). The Question Generation Module receives this information and interacts with a Large Language Model (LLM) in order to generate relevant interview questions [18]. Candidate responses will go through two parallel data processing paths: 1) Text Evaluation Path, 2) Speech Analysis Path. Results from these two paths are consolidated in order to provide a user with final feedback. Details of how the system architecture was developed are presented in Fig 1.

System Workflow Flowchart

The flow of the system illustrated in Fig. 2 begins when a candidate selects a topic to be interviewed on or uploads a résumé. From there, the system creates interview questions based on the selection and captures the response of the candidate. If the candidate's response is in audio format, the audio response will be converted to text using speech recognition technology. The transcription will then be

checked for accuracy with the use of the answer provided by the candidate. The audio signal, when being assessed for accuracy, is put through a process called feature extraction. Feature extraction generates information that is used by a machine learning algorithm to generate a confidence score. Finally, all outputs from the evaluation process will be combined into structured feedback provided to the candidate.

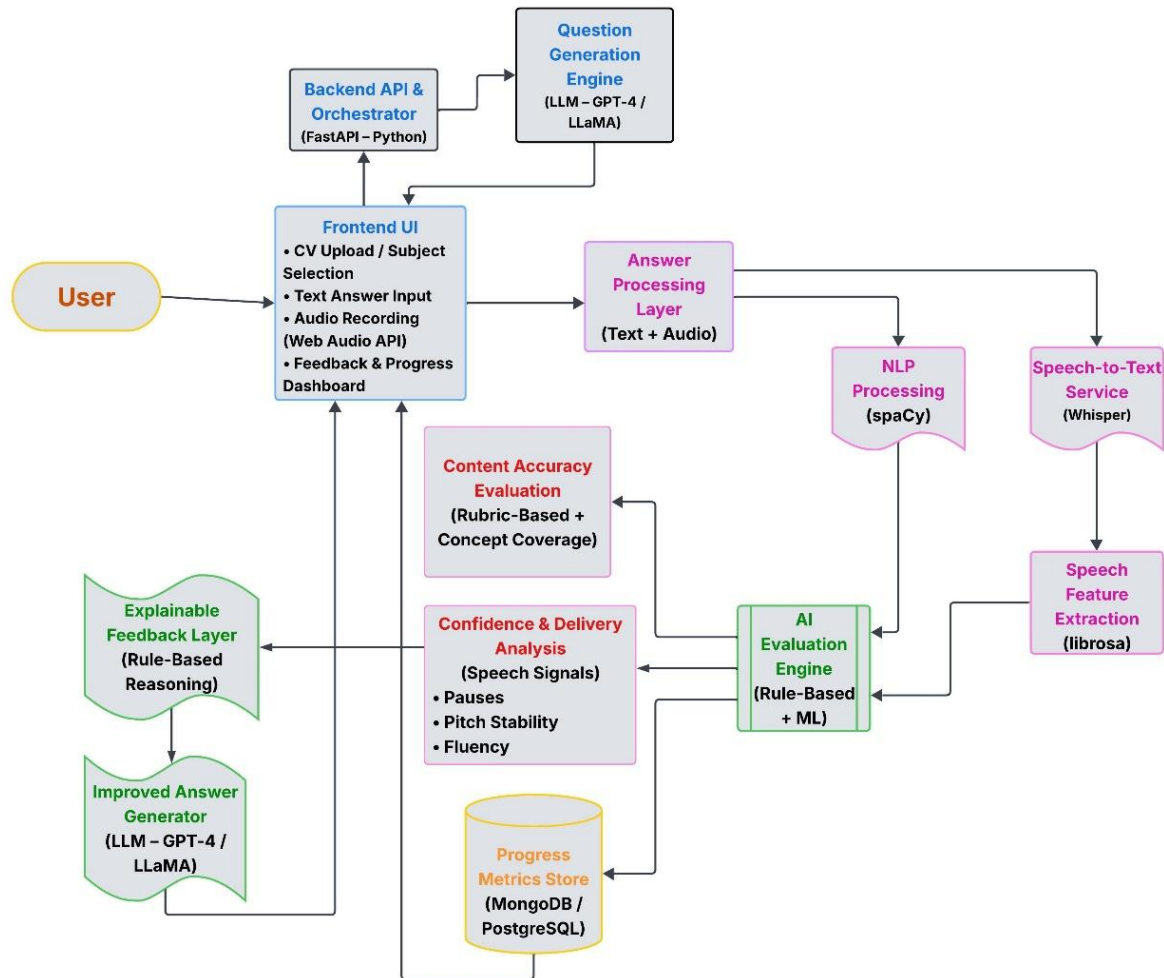


Fig. 1 Architecture of the system

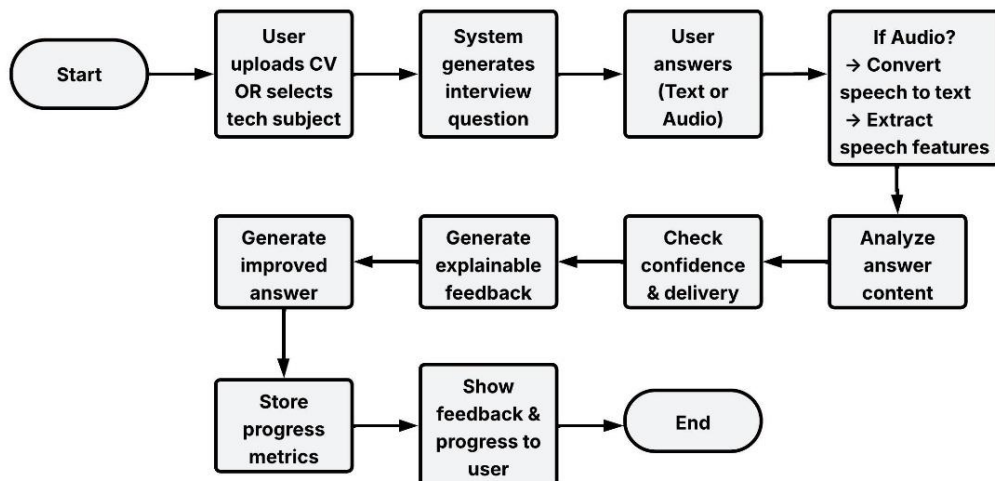


Fig.2 Flowchart of System

IV. RESULTS

4.1 Dataset Description

Because of the limited availability of publicly accessible labelled interview data, we generated a synthetic interview dataset comprising 10763 rows of acoustic features sampled from realistic statistical distributions. For example, pitch values were sampled evenly from traditional ranges for males and females, and the number of filler words used followed a Poisson distribution. To label the samples beyond a linear combination of features, we used Llama 3.1 through the Groq API for holistic labelling.

The LLM processed batches of feature data and assigned each sample a confidence score and reasoning string based off complex interactions that would not be apparent using a rule-based formula. After an extensive process of cleaning data and eliminating outliers, the final training set was composed of 8982 samples.

4.2 Data Visualization

Visualizing data to help understand the distribution and relationships between speech features. Multiple visualizations were created to analyze the dataset and its correlations with the confidence score.

Fig.3 shows Histograms illustrating the frequency distributions for the nine acoustic features, after data cleaning

and removal of outliers. Observations include the bimodal nature of pitch_mean, which represents the natural variation in pitch from male to female, and the “long-tail” distribution of both filler_words and pause_count, which exhibit Poisson-like distribution characteristics. The distribution patterns of speech features demonstrate that they capture realistic ranges of human speech, from highly fluent to moderately fluent to non-fluent delivery.

As shown in Fig. 4, the training set, validation set, and test set all have very similar target confidence label distributions when depicted in histogram form. In fact, the means (μ) are fairly stable, ranging from 5.31 to 5.57, across all three data partitions. This stability shows that evaluations conducted on the model were performed with a representative subgroup of the entire population.

Fig.5 shows that energy_mean has the highest and strongest positive correlation ($r = 0.863$) to confidence, indicating that loudness of voice is a primary indicator of confidence. Jitter ($r = -0.763$) and filler_words ($r = -0.753$) have the highest and strongest negative correlations with confidence and support the theory that vocal instability and use of filler words are indicators of poor delivery. Additionally, the ranking of feature importance shows that 50% of the model's classification decisions are based on the volume of voice; followed by fluency and pitch stability.

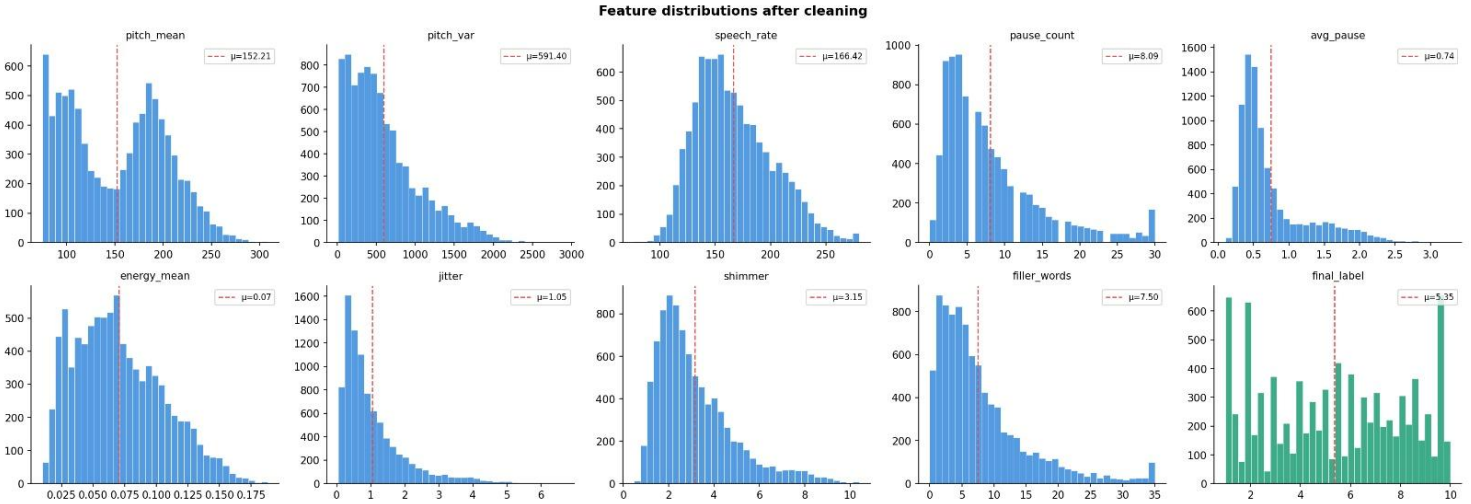


Fig. 2 Histograms of acoustic features

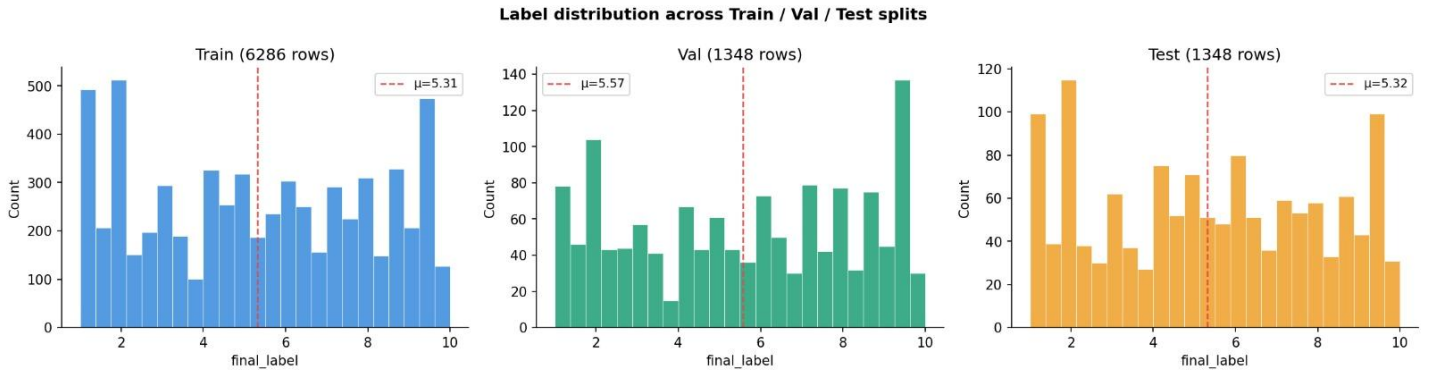


Fig. 4 Label Distribution across Train/Val/Test splits

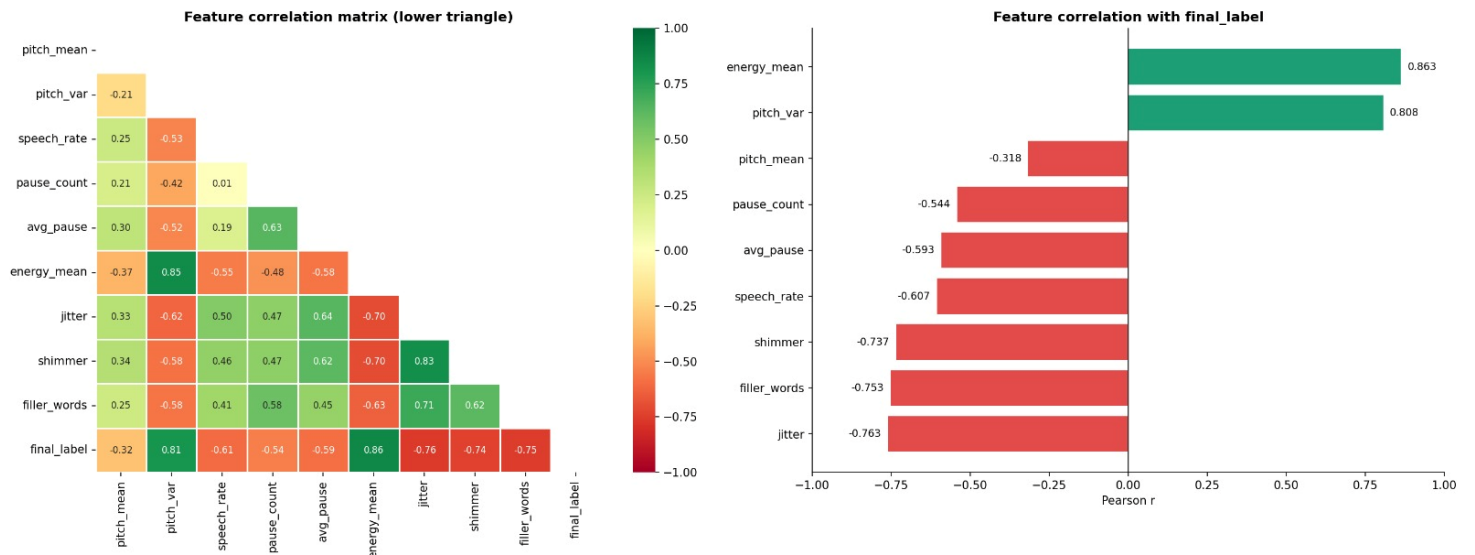


Fig. 5 Feature correlation matrix

4.3 Model Comparison

The performance evaluation of a wide range of different types of machine learning models was run using speech feature data to find the best way to model confidence score predictions from these speech features. All models were created using the same dataset, and performance was evaluated based on standard metrics. Table 1 shows the performance comparison for all of the models.

Table 1: Model Performance Comparison

Model	MAE	RMSE	R ²
XGBoost	0.464	0.715	0.932
MLP	0.472	0.714	0.932
Random Forest	0.477	0.733	0.928
Ridge Regression	0.676	0.918	0.887

For the XGBoost algorithm alone, the R2 score of this model = 0.932 indicates that the model can account for 93.2% of the variation in confidence score data. In addition, the MAE of the predicted value (0.464) indicates that on average, the model predictions are within one-half of a point from the actual value of confidence score when confidence scores are measured on a 1-10 scale.

4.4 Feature Importance Analysis

One of the benefits of using XGBoost is that it provides an indication of feature importance value based on their contribution toward making predictions with the model. Feature importance analysis indicated that the following four features were major contributors in predicting the confidence score of the esteem of others' confidence in a speaker's verbal delivery:

- Speech Rate
- Pause Frequency
- Pitch Variation
- Energy Level
- Filler Word Frequency

There were two indications regarding how hesitation through pause counts and use of filler words with respect to confidence scores appeared in the analysis, i.e., that lack of hesitation through bountiful rates of pause and filler words yielded a strong negative relationship to confidence scores; however, high consistency in speech rates and stable pitch yielded high confidence score predictions. Using these indicators provides useful means of explainable feedback to users of their speaking performance.

V. CONCLUSION

This Interview Coaching System uses two separate pipelines (Content Evaluation using Gemini LLM and Delivery Confidence using XGBoost) to simulate an interview experience. In doing so, candidates receive feedback not only on the factual accuracy of their answers but also on their use of verbal/vocal indicators (e.g., volume and pitch stability) indicating confidence.

Thus, the system offers transparency and actionable metrics via the application of Explainable AI (XAI) principles. Functioning within the candidates as a result of the high accuracy of the XGBoost model (with an R² of 0.932) and considering that acoustic features indicated as good indicators of perceived confidence include: energy mean, jitter and filler words, it can be inferred that there is a significant amount of reliability to these specific features. The results also support future development to allow for the implementation of on-line/synthetic and real-time feedback and support for multiple languages, so as to help job seekers pursue success in a global job market during this competitive time.

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